

Independent Study: Cosmin Ilut

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Abstract

This document provides additional explorations of draft framework proposed in Ilut Valchev working paper.

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1 Setup

We consider an agent with an expected utility function facing a consumption problem over a sequence of periods. At each period t , the agent chooses consumption c_t , and perfectly observes state variable s_t . The utility function is a function of c_t and s_t and therefore the agent perfectly observes their payoff $u(c_t, s_t)$ at each period. The agent discounts future utility by a factor $\beta \in (0, 1)$.

$$V^*(s_t) = \max_{c_t \in B(s_t)} \{u(c_t, s_t) + \beta \mathbb{E}[V^*(s_{t+1})]\} \quad (1)$$

This is the Bellman equation for the agent's state value function $V^*(s_t)$, where $B(s_t)$ is the agent's budget set at state s_t . This is directly linked to the agent's action value function $Q^*(s_t, c_t)$ as follows¹.

$$Q^*(s_t, c_t) = u(c_t, s_t) + \beta \mathbb{E}[V^*(s_{t+1})] \quad (2)$$

There is a stochastic Markov transition function $\mathcal{F}(c_t, s_t, \epsilon_{t+1})$ where $\epsilon_{t+1} \sim \mathcal{G}(\epsilon_{t+1})$ is an i.i.d. shock that perturbs the transition from s_t to s_{t+1} .

1.1 Agent's Beliefs

The agent reasonably believes there to be some unknown optimal policy π^* that produces optimal state action value function $Q^*(s_t, c_t)$. As the agent has not learned this policy yet, they have a belief over the set of possible action value functions $\mathcal{Q}$, which are modeled as Gaussian Processes (GPs).

$$Q^*(c, s) \sim GP\left(\hat{Q}_0(c, s), \hat{\Sigma}_0((s, c), (s', c'))\right) \quad (3)$$

The mean function is $\hat{Q}_0(c, s)$ and the covariance function is $\hat{\Sigma}_0((s, c), (s', c'))$. The use of GPs becomes useful in that they allow the agent to have correlated beliefs over similar state-action pairs, allowing for generalization across the space $\mathcal{Q}$. And, as new observations are made, the agent can update their beliefs using Bayesian updating, resulting in a new GP posterior over $\mathcal{Q}$.

1.1.1 Useful Properties of GPs

There are some useful properties of GPs that will list out here for reference.

Useful Fact 1 (Closure Under Conditioning) *If $f \sim GP(\mu, k)$ and we have observations $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ where $y_i = f(x_i) + \epsilon_i$ with $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$. The posterior distribution*

¹Here we are using $*$ to indicate the state and action value functions under the optimal policy, there is no difference in the formation of the equations under different policies. Notationally, it would be indicated as V^π and Q^π for a policy π .

$f|\mathcal{D} \sim GP(\mu_n, k_n)$ where

$$\begin{aligned}\mu_n(x) &= \mu(x) + k(x, X_n)(k(X_n, X_n) + \sigma^2 I)^{-1}(Y_n - \mu(X_n)) \\ k_n(x, x') &= k(x, x') - k(x, X_n)(k(X_n, X_n) + \sigma^2 I)^{-1}k(X_n, x')\end{aligned}$$

where $X_n = (x_1, \dots, x_n)^\top$ is the matrix of observed inputs and $Y_n = (y_1, \dots, y_n)^\top$ is the vector of observed outputs.

Useful Fact 2 (Finite-Dimensional Marginals Are Gaussian) *For any finite collection of points $\{x_1, \dots, x_k\}$, the joint distribution $(f(x_1), \dots, f(x_k))$ is multivariate normal. This implies that predictions at new points have joint Gaussian distributions.*

Useful Fact 3 (Linearity of the Mean) *If $f \sim GP(\mu, k)$, then for any linear functional L :*

$$L[f] \sim \mathcal{N}(L[\mu], L[k])$$

This extends to integration against measures. For instance, if the agent needs $\mathbb{E}_{s'}[V^(s')] = \mathbb{E}_{s'}[\max_c Q^*(s', c)]$, the inner expectation over the transition kernel is tractable. This is because the expectation operator defines a bounded linear functional on $L^1(\Omega, \mathcal{F}, P)$. though the max operator introduces nonlinearity.*

1.1.2 The Negative Exponential and an Example

A common choice for the covariance function is the squared exponential (or radial basis function) kernel:

$$\hat{\Sigma}_0((s, c), (s', c')) = \sigma_0^2 \exp(-\psi_s \cdot \|s - s'\| - \psi_c \cdot \|c - c'\|) \quad (4)$$

where σ_0^2 is the signal variance, and ψ_s, ψ_c degree of correlation or smoothness across states and actions respectively. If ψ_s is large, then the agent believes that small changes in state will lead to large changes in the action value function, indicating a less smooth function in states. If ψ_c is small, then the agent believes that changes in action will have little effect on the action value function, indicating a smoother function in actions. Here $\|\cdot\|$ is the norm on the state and action spaces respectively.

Example. Consider a two-period consumption-savings problem with log utility where wealth is the state variable. In period 1, the agent has wealth s and chooses consumption c . The remaining wealth is saved and grows at rate R to become wealth s' in period 2, where the agent consumes everything.

- **Period 1:** Agent has wealth s , chooses consumption $c \in [0, s]$, and saves $s - c$
- **Period 2:** Wealth becomes $s' = R(s - c)$ with $R = 1.1$, and the agent consumes everything

True Q^* : The agent receives utility today plus discounted utility tomorrow: For instance, if $s = 10$ and $c = 4$ then $Q^*(10, 4) = \log(4) + 0.9 \log(6.6) \approx 1.39 + 1.70 = 3.09$. The agent does not know this function. They place a GP prior with mean zero and exponential kernel with $\sigma_0^2 = 1$, $\psi_s = 0.5$, $\psi_c = 0.5$. The agent tries $(s, c) = (10, 4)$ and observes $y_1 = 3.09$. At a nearby point like $(10, 5)$, the posterior mean is pulled toward 3.09 because the kernel implies these points are correlated:

$$\hat{\Sigma}((10, 4), (10, 5)) = \exp(-0.5 \cdot 0 - 0.5 \cdot 1) = e^{-0.5} \approx 0.61$$

At a distant point like $(2, 1)$:

$$\hat{\Sigma}((10, 4), (2, 1)) = \exp(-0.5 \cdot 8 - 0.5 \cdot 3) = e^{-5.5} \approx 0.004$$

The correlation is negligible, so the agent learns almost nothing about $(2, 1)$ from the observation at $(10, 4)$.

1.1.3 The Distribution of Belief Over Optimal Policy

The agent's prior is conveniently expressed as a gaussian process over action value functions. This allows the agent to perform Bayesian updating as new observations are made. Perhaps the more natural construction would be to have an agent to have a belief over policies π . When the agent enters a new environment they have some belief over what actions to take under various conditions (states). They revise their beliefs as they observe new outcomes.

As we know from the RL theory, the optimal policy, optimal value function, and optimal action value functions are intrinsically linked.

$$\pi^*(s) = \arg \max_c Q^*(s, c), \quad \text{where } Q^*(s, c) = \sup_{\pi} Q^{\pi}(s, c) \quad (5)$$

$$V^*(s) = \max_c Q^*(s, c), \quad \text{where } V^*(s) = \sup_{\pi} V^{\pi}(s) \quad (6)$$

Given the definitions are circular we can convert between them as follows:

$$\textbf{From } Q^* \textbf{ to } \pi^*: \quad \pi^*(s) = \arg \max_{c \in B(s)} Q^*(s, c) \quad (7)$$

$$\textbf{From } Q^* \textbf{ to } V^*: \quad V^*(s) = \max_{c \in B(s)} Q^*(s, c) = Q^*(s, \pi^*(s)) \quad (8)$$

$$\textbf{From } V^* \textbf{ to } Q^*: \quad Q^*(s, c) = u(c, s) + \beta \mathbb{E}[V^*(s') | s, c] \quad (9)$$

$$\textbf{From } \pi^* \textbf{ to } V^*: \quad V^*(s) = u(\pi^*(s), s) + \beta \mathbb{E}[V^*(s') | s, \pi^*(s)] \quad (10)$$

To compute the agent's prior belief over policies π , we need to calculate the distribution of the arg max of a GP. This is not analytically tractable: if $f \sim GP(\mu, k)$, then $\arg \max_x f(x)$

is a random element of the input space whose distribution depends on the entire sample path structure and has no closed form.

1.2 Learning Process

The agent updates her beliefs over $\mathcal{Q}$ through two mechanisms: experience and reasoning. The experience mechanism involves updating beliefs through direct observation of realized outcomes (s_t, c_t, u_t, s_{t+1}) from actions taken. The reasoning mechanism involves updating beliefs through mental simulation—evaluating counterfactual actions using the agent’s model of transitions $\mathcal{F}$ and payoffs u without taking those actions.

1.2.1 Experience-Based Learning

The agent takes a consumption action c_{t-1} in state s_{t-1} , receives payoff $u_{t-1} = u(c_{t-1}, s_{t-1})$, and transitions to state s_t , based on the Markov transition function $\mathcal{F}(c_{t-1}, s_{t-1}, \epsilon_t)$. They can now look back and form a temporal difference (TD) approximation. The Bellman equation may be rearranged as follows:

$$\underbrace{u(c_{t-1}, s_{t-1})}_{\text{observed}} = \underbrace{Q^*(c_{t-1}, s_{t-1})}_{\text{unknown, GP prior}} - \beta \underbrace{\mathbb{E}_{t-1}[Q^*(s_t, \pi^*(s_t))]}_{\text{requires integration over } \mathcal{F}} \quad (11)$$

The agent approximates this by replacing the expectation with the realized value:

$$\underbrace{u(c_{t-1}, s_{t-1})}_{\text{observed}} \approx \underbrace{Q^*(c_{t-1}, s_{t-1})}_{\text{unknown, GP prior}} - \beta \underbrace{Q^*(s_t, \tilde{c}_t)}_{\text{unknown, GP prior}} \quad (12)$$

where $\tilde{c}_t = \arg \max_{c \in B(s_t)} \hat{Q}_{t-1}(c, s_t)$ is the greedy action under current beliefs and s_t is the realized state. The agent can then perform a posterior update on their GP prior. Define the experience signal:

$$\eta_t^E \equiv u(c_{t-1}, s_{t-1}) \quad (13)$$

This signal is a noisy observation of the linear functional $Q^*(c_{t-1}, s_{t-1}) - \beta Q^*(s_t, \tilde{c}_t)$. Since linear functionals of a GP are Gaussian, standard conditioning applies. The posterior mean updates as:

$$\hat{Q}_t^E(c, s) = \hat{Q}_{t-1}(c, s) + \alpha_t^E(c, s) \left[\eta_t^E - \left(\hat{Q}_{t-1}(c_{t-1}, s_{t-1}) - \beta \hat{Q}_{t-1}(s_t, \tilde{c}_t) \right) \right] \quad (14)$$

where the term in brackets is the TD error—the discrepancy between observed utility and predicted utility under current beliefs. The gain $\alpha_t^E(c, s)$ is the Kalman gain:

$$\alpha_t^E(c, s) = \frac{\text{Cov}(\eta_t^E, Q^*(c, s) \mid \mathcal{I}_{t-1})}{\text{Var}(\eta_t^E \mid \mathcal{I}_{t-1})} \quad (15)$$

This determines how much the belief at (c, s) adjusts per unit of TD error, based on the kernel-implied correlation with the observed state-action pairs. The posterior covariance shrinks accordingly:

$$\hat{\Sigma}_t^E(c, s, c', s') = \hat{\Sigma}_{t-1}(c, s, c', s') - \alpha_t^E(c, s) \cdot \text{Cov}(\eta_t^E, Q^*(c', s') \mid \mathcal{I}_{t-1}) \quad (16)$$

This update works because the experience signal is a linear functional of the GP. The agent observes $\eta_t^E = u(c_{t-1}, s_{t-1})$, which they interpret as an observation of $Q^*(c_{t-1}, s_{t-1}) - \beta Q^*(s_t, \tilde{c}_t)$ which is a linear combination of Q^* evaluated at two points. Since linear functionals of a GP are Gaussian, and conditioning a joint Gaussian on one component yields a Gaussian posterior, the updated beliefs remain a GP.

Noise in Experience Signal Because the experience signal is based on a single realized transition, it is subject to noise from the stochastic transition function $\mathcal{F}$. As such, agents may misattribute actions to outcomes due to random shocks. This can cause posterior updates that updates upwards or downwards incorrectly.

1.2.2 An Example of Experience-Based Learning

We illustrate the experience-based learning mechanism with a tractable consumption-savings example.

Environment. Consider an agent with quadratic utility $u(c) = c - \frac{\alpha}{2}c^2$ facing a linear wealth transition $s_{t+1} = R(s_t - c_t) + y_t$, where $y_t \in \{y_L, y_M, y_H\}$ is income drawn uniformly, so $\mathbb{E}[y_t] = y_M$. We impose $\beta R = 1$, which implies the optimal policy is to smooth consumption: $c^* = \frac{R-1}{R}s + y_M$.

Functional Form of Q^* . With quadratic utility and linear transitions, the Bellman equation admits a polynomial solution. Therefore:

$$Q^*(s, c) = \theta_0 + \theta_1 s + \theta_2 c + \theta_3 c^2 + \theta_4 s c + \theta_5 s^2 \quad (17)$$

where the coefficients $\boldsymbol{\theta} = (\theta_0, \theta_1, \theta_2, \theta_3, \theta_4, \theta_5)^\top$ depend on the primitives $\alpha, \beta, R, y_M, \text{Var}(y)$.

Agent's Prior. The agent knows Q^* is polynomial in (s, c) but does not know the coefficients. They hold a Gaussian prior:

$$\boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) \quad (18)$$

Define the feature vector $\boldsymbol{\phi}(s, c) = (1, s, c, c^2, s c, s^2)^\top$, so that $Q^*(s, c) = \boldsymbol{\phi}(s, c)^\top \boldsymbol{\theta}$.

Greedy Policy. Given current beliefs $\hat{\boldsymbol{\mu}}_{t-1}$, the agent chooses:

$$c_t = \arg \max_c \boldsymbol{\phi}(s_t, c)^\top \hat{\boldsymbol{\mu}}_{t-1} \quad (19)$$

Bayesian Updating. The agent observes utility $\eta_t^E = u(c_{t-1})$ and interprets it via the TD approximation:

$$\eta_t^E = Q^*(s_{t-1}, c_{t-1}) - \beta Q^*(s_t, \tilde{c}_t) = \mathbf{x}_t^\top \boldsymbol{\theta} \quad (20)$$

where $\mathbf{x}_t = \boldsymbol{\phi}(s_{t-1}, c_{t-1}) - \beta \boldsymbol{\phi}(s_t, \tilde{c}_t)$ and $\tilde{c}_t$ is the greedy action at s_t . The posterior update is:

$$\hat{\boldsymbol{\mu}}_t = \hat{\boldsymbol{\mu}}_{t-1} + \mathbf{K}_t \left(\eta_t^E - \mathbf{x}_t^\top \hat{\boldsymbol{\mu}}_{t-1} \right) \quad (21)$$

$$\hat{\boldsymbol{\Sigma}}_t = \hat{\boldsymbol{\Sigma}}_{t-1} - \mathbf{K}_t \mathbf{x}_t \mathbf{x}_t^\top \hat{\boldsymbol{\Sigma}}_{t-1} \quad (22)$$

where the Kalman gain is:

$$\mathbf{K}_t = \frac{\hat{\boldsymbol{\Sigma}}_{t-1} \mathbf{x}_t}{\mathbf{x}_t^\top \hat{\boldsymbol{\Sigma}}_{t-1} \mathbf{x}_t} \quad (23)$$

Misattribution from Income Shocks. Suppose the agent experiences income sequence $y_L, y_M, y_M, y_H, y_M, y_M$. In period 1, income $y_L < y_M$ realizes. The agent arrives at state $s_1 = R(s_0 - c_0) + y_L$, which is lower than expected. The TD error $\eta_1^E - \mathbf{x}_1^\top \hat{\boldsymbol{\mu}}_0$ is negative: the agent received less than predicted. They update $\hat{\boldsymbol{\mu}}_1$ downward, concluding that consuming c_0 at wealth s_0 was worse than previously believed. But the bad outcome came from unlucky income, not from the consumption choice. The agent misattributes luck to policy.

When y_H realizes in period 4, the opposite occurs: the agent arrives at a higher state than expected, the TD error is positive, and they update upward—again misattributing the income shock to their action. Over the sequence, beliefs oscillate with income realizations rather than converging smoothly to the truth.

1.2.3 Learning Through Reasoning

Reasoning-based learning occurs in a reduced-form manner. We suppose that the agent can produce mental simulations and evaluate counterfactual actions, developing a noisy vector of signals $\boldsymbol{\eta}_t^R$ that provide information about the action value function. To formalize this, fix the current state s_t and consider the vector of action values across all possible consumption levels:

$$\mathbf{Q}^*(s_t) = (Q^*(s_t, c_1), Q^*(s_t, c_2), \dots, Q^*(s_t, c_n))^\top \quad (24)$$

By the GP assumption, this vector is jointly Gaussian with mean $\hat{\mathbf{Q}}_{t-1}(s_t)$ and covariance $\hat{\boldsymbol{\Sigma}}_{t-1}(s_t)$ derived from the kernel.

The agent generates k reasoning signals by taking linear combinations of this vector:

$$\boldsymbol{\eta}_t^R = \boldsymbol{\Omega}_t' \mathbf{Q}^*(s_t) + \boldsymbol{\epsilon}_{\eta,t}, \quad \boldsymbol{\epsilon}_{\eta,t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\eta,t}) \quad (25)$$

where $\mathbf{\Omega}_t$ is an $n \times k$ loadings matrix describing which comparisons the agent evaluates. Each column of $\mathbf{\Omega}_t$ specifies one signal. As an example, to evaluate the action value of consuming $\tilde{c}_j$, set column i to have a 1 in row j and zeros elsewhere. The signal is $Q^*(s_t, \tilde{c}_j) + \epsilon$. To compare actions $\tilde{c}_j$ versus $\tilde{c}_k$, set column i to have +1 in row j , -1 in row k , and zeros elsewhere. The signal is $Q^*(s_t, \tilde{c}_j) - Q^*(s_t, \tilde{c}_k) + \epsilon$. The noise covariance $\mathbf{\Sigma}_{\eta,t}$ reflects the precision of the agent's mental simulation.

The agent can use this set of reasoning signals to update their beliefs about Q^* , similar to the experience-based update. The resulting posterior mean is:

$$\hat{Q}_t^R(c, s) = \hat{Q}_{t-1}(c, s) + \boldsymbol{\alpha}_t^R(c, s)^\top \left[\boldsymbol{\eta}_t^R - \mathbf{\Omega}_t' \hat{Q}_{t-1}(s_t) \right] \quad (26)$$

$$\boldsymbol{\alpha}_t^R(c, s) = \text{Cov}(\boldsymbol{\eta}_t^R, Q^*(c, s) \mid \mathcal{I}_{t-1}) \left(\text{Var}(\boldsymbol{\eta}_t^R \mid \mathcal{I}_{t-1}) \right)^{-1} \quad (27)$$

1.3 Action Selection

At each period t , the agent goes through a set of steps: they first perform a bayesian update of their beliefs based on realized utility from the previous period (experience-based learning). They then must choose how much reasoning to do, they then update their beliefs based on the reasoning signals. Finally, they choose an action based on their updated beliefs constrained by both their budget set and an exploration mechanism.

There are also, therefore, two versions of the payoff function. The actual payoff function, which is unknown to the agent and defined under the true optimal policy π^* , is $Q^*(s, c)$. The believed payoff function after reasoning-based learning is $\hat{Q}_t^R(s, c)$.

$$\text{Expected Actual Payoff:} \quad \mathbb{E} \left[\sum_c Q^*(c, s_t) \cdot \pi_t(c|s_t) \mid \mathcal{I}_t^E \right] \quad (28)$$

$$\text{Expected Believed Payoff:} \quad \mathbb{E} \left[\sum_c \hat{Q}_t^E(c, s_t) \cdot \pi_t(c|s_t) \mid \mathcal{I}_t^E \right] \quad (29)$$

1.3.1 The Reasoning Decision

The agent chooses the simulations to run ($\mathbf{\Omega}_t$) and the precision of those simulations ($\mathbf{\Sigma}_{\eta,t}$). They maximize the expected payoff net of reasoning costs:

$$\max_{\mathbf{\Omega}_t, \mathbf{\Sigma}_{\eta,t}} \mathbb{E} \left[\sum_c \hat{Q}_t^R(c, s_t) \cdot \pi_t(c|s_t) \mid \mathcal{I}_t^E \right] - \kappa \ln \left[\frac{|\hat{\mathbf{\Sigma}}_t^E|}{|\hat{\mathbf{\Sigma}}_t^R|} \right] \quad (30)$$

The cost term is the information gain from reasoning, measured as the entropy reduction in beliefs about Q^* . The lower the posterior covariance $\hat{\mathbf{\Sigma}}_t^R$, the more precise the beliefs, and the higher the cost. The form is motivated by rational inattention models (Sims 2003, Matejka and McKay 2015).

1.3.2 The Action Decision

The agent’s action decision is governed by their policy function $\pi_t(c|s_t)$. This policy is derived from maximizing expected payoffs subject to exploration constraints:

$$\max_{\pi_t(c|s_t)} \sum_c \hat{Q}_t^R(c, s_t) \cdot \pi_t(c|s_t) \quad (31)$$

$$\text{subject to } - \sum_c \pi_t(c|s_t) \ln \pi_t(c|s_t) \geq h \sum_c \sigma_t(c, s_t) \quad (32)$$

The constraint requires that the entropy of the policy exceeds a threshold scaled by the agent’s total posterior uncertainty across actions, ensuring greater exploration when estimates are imprecise. The solution to this constrained optimization yields the following action distribution:

$$\pi_t(c|s_t) = \begin{cases} \frac{\exp\left(\frac{\hat{Q}_t^R(c, s_t)}{\delta_t}\right)}{\sum_{c'} \exp\left(\frac{\hat{Q}_t^R(c', s_t)}{\delta_t}\right)} & \text{if } \delta_t > 0 \\ c = \arg \max_{c'} \hat{Q}_t^R(c', s_t) & \text{if } \delta_t = 0 \end{cases} \quad (33)$$

where δ_t is the Lagrange multiplier on the entropy constraint.

A note on the policy function. The policy function is essential for determining optimal reasoning in 1.3.1, as the expected payoff from reasoning depends on the action distribution. Consequently, the agent solves the action decision problem twice: first, after experience-based learning to derive $\pi_t^E(c|s_t)$ for reasoning, and second, after reasoning-based learning to determine $\pi_t^R(c|s_t)$ for action selection.

2 Investment Problem

Consider a single firm's (j) investment problem. The firm has a production function $f(z_{j,t}, k_{j,t})$, where $z_{j,t}$ is an idiosyncratic productivity shock and $k_{j,t}$ is the capital stock. The firm chooses investment $i_{j,t}$ to maximize the expected discounted value of future dividends, which are subject to adjustment costs $\phi(i_{j,t}, k_{j,t})$. Capital evolves according to a standard law of motion with depreciation δ :

$$\max_{\{i_{j,t}\}_{t=0}^{\infty}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(d_{j,t}) \right] \quad (34)$$

$$\text{s.t.} \quad d_{j,t} = f(z_{j,t}, k_{j,t}) - i_{j,t} [1 + \phi(i_{j,t}, k_{j,t})] \quad (35)$$

$$k_{j,t+1} = (1 - \delta)k_{j,t} + i_{j,t} \quad (36)$$

2.1 Simplifications

This problem can be simplified in several steps. First, we assume utility is linear in dividends, $u(d_{j,t}) = d_{j,t}$, and adjustment costs are linear, $\frac{\varkappa}{2} \frac{i_{j,t}^2}{k_{j,t}}$, while ignoring idiosyncratic shocks. In the next step, we reintroduce idiosyncratic shocks that follow an AR(1) process. Subsequently, we incorporate non-linear utility. Finally, we consider quadratic adjustment costs.

2.1.1 Base Case - Linear Utility, No Shocks, Linear Adjustment Costs

In the base case, the investment problem for firm j can be expressed using the following Bellman equation (dropping the firm subscript j for simplicity):

$$v(k_t) = \max_{i_t} \left\{ k_t^\alpha - i_t \left[1 + \frac{\varkappa}{2} \frac{i_t}{k_t} \right] + \beta v(k_{t+1}) \right\}$$

$$\text{s.t.} \quad k_{t+1} = (1 - \delta)k_t + i_t$$

This results in the following conditions and Euler equation:

$$\begin{aligned} \text{FOC:} \quad & \beta \frac{dv}{dk}(k_{t+1}) = 1 + \frac{\varkappa i_t}{k_t} \\ \text{Envelope:} \quad & \frac{dv}{dk}(k_t) = \alpha k_t^{\alpha-1} + \frac{\varkappa i_t^2}{2k_t^2} + \beta \frac{dv}{dk}(k_{t+1})(1 - \delta) \\ \text{Euler:} \quad & \frac{1}{\beta} \left(1 + \frac{\varkappa i_t}{k_t} \right) = \alpha k_{t+1}^{\alpha-1} + \frac{\varkappa i_{t+1}^2}{2k_{t+1}^2} + (1 - \delta) \left(1 + \frac{\varkappa i_{t+1}}{k_{t+1}} \right) \end{aligned}$$

2.1.2 Introducing Idiosyncratic Shocks

We now introduce idiosyncratic productivity shocks z_t that follow an AR(1) process in logs.

$$\ln(z_{t+1}) = \rho \ln(z_t) + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, \sigma_z^2) \quad (37)$$

The Bellman equation now becomes:

$$\begin{aligned} v(z_t, k_t) = \max_{i_t} & \left\{ z_t k_t^\alpha - i_t \left[1 + \frac{\varkappa}{2} \frac{i_t}{k_t} \right] + \beta \mathbb{E}_{z_{t+1}|z_t} [v(z_{t+1}, k_{t+1})] \right\} \\ \text{s.t.} \quad & k_{t+1} = (1 - \delta)k_t + i_t \end{aligned}$$

This leads to the following conditions and Euler equation:

$$\begin{aligned} \textbf{FOC:} \quad & \beta \mathbb{E}_{z_{t+1}|z_t} \left[\frac{\partial v}{\partial k}(z_{t+1}, k_{t+1}) \right] = 1 + \frac{\varkappa i_t}{k_t} \\ \textbf{Envelope:} \quad & \frac{\partial v}{\partial k}(z_t, k_t) = z_t \alpha k_t^{\alpha-1} + \frac{\varkappa i_t^2}{2k_t^2} + \beta \mathbb{E}_{z_{t+1}|z_t} \left[\frac{\partial v}{\partial k}(z_{t+1}, k_{t+1}) \right] (1 - \delta) \\ \textbf{Euler:} \quad & \frac{1}{\beta} \left(1 + \frac{\varkappa i_t}{k_t} \right) = \mathbb{E}_{z_{t+1}|z_t} \left[z_{t+1} \alpha k_{t+1}^{\alpha-1} + \frac{\varkappa i_{t+1}^2}{2k_{t+1}^2} + (1 - \delta) \left(1 + \frac{\varkappa i_{t+1}}{k_{t+1}} \right) \right] \end{aligned}$$

2.1.3 Non-Linear Utility

We suppose the firm is risk-averse and has logarithmic utility over dividends, $u(d_{j,t}) = \ln(d_{j,t})$. The Bellman equation is now:

$$\begin{aligned} v(z_t, k_t) = \max_{i_t} & \left\{ \ln \left(z_t k_t^\alpha - i_t \left[1 + \frac{\varkappa}{2} \frac{i_t}{k_t} \right] \right) + \beta \mathbb{E}_{z_{t+1}|z_t} [v(z_{t+1}, k_{t+1})] \right\} \\ \text{s.t.} \quad & k_{t+1} = (1 - \delta)k_t + i_t \end{aligned}$$

Define dividends as $d_t \equiv z_t k_t^\alpha - i_t \left[1 + \frac{\varkappa}{2} \frac{i_t}{k_t} \right]$. This results in the following conditions and Euler equation:

$$\begin{aligned} \textbf{FOC:} \quad & \beta \mathbb{E}_{z_{t+1}|z_t} \left[\frac{\partial v}{\partial k}(z_{t+1}, k_{t+1}) \right] = \frac{1}{d_t} \left(1 + \frac{\varkappa i_t}{k_t} \right) \\ \textbf{Envelope:} \quad & \frac{\partial v}{\partial k}(z_t, k_t) = \frac{1}{d_t} \left(z_t \alpha k_t^{\alpha-1} + \frac{\varkappa i_t^2}{2k_t^2} \right) + \beta \mathbb{E}_{z_{t+1}|z_t} \left[\frac{\partial v}{\partial k}(z_{t+1}, k_{t+1}) \right] (1 - \delta) \\ \textbf{Euler:} \quad & \frac{1}{\beta} \left(1 + \frac{\varkappa i_t}{k_t} \right) = \mathbb{E}_{z_{t+1}|z_t} \left[\frac{d_t}{d_{t+1}} \left(z_{t+1} \alpha k_{t+1}^{\alpha-1} + \frac{\varkappa i_{t+1}^2}{2k_{t+1}^2} + (1 - \delta) \left(1 + \frac{\varkappa i_{t+1}}{k_{t+1}} \right) \right) \right] \end{aligned}$$